

Rexy Bridge 3.0 — User Guide

Print / greyscale edition — light background, mono-friendly. For the on-screen colour version see [USER_GUIDE.pdf](#).

A panel-by-panel tour of every control, in the order they appear in the app. Setup and UE connection are covered in **GETTING_STARTED.md**; this guide assumes you're connected and driving a camera.



v3.0.0-beta.1

REXYWHEELS
DEVICEWS
CONNECTEDUE —
waiting...

UNITS

M /
CMFT
/ IN

OUTPUT

OSC
/ UE

FREED

SCRIPT HOST

127.0.0.1

WS PORT

9000

UES HOST

127.0.0.1

UES OSC PORT

8000

BASE PATH

/rexy

CONFIG

EXPORT

IMPORT

ACTIVE CAMERA

— none found - expose a property per camera —

APPLY BINDINGS TO ALL

SAVE BINDINGS

SCAN UE

REXY WHEELS — CAMERA HEAD

ABSOLUTE

RELATIVE

Pan /rexy/pan

+96.6°

live —

PAD1-AX3 HOLD INV RESET

S 1. C 1. D 0 F 0
MIN (°) -180 MAX (°) 180

Tilt /rexy/tilt

+0.0°

live —

A / D HOLD INV RESET

S 0. C 1. D 0 F 16
MIN (°) -90 MAX (°) 90

Roll /rexy/roll

+0.0°

live —

BIND HOLD INV RESET

S 1. C 1. D 0 F 0
MIN (°) -45 MAX (°) 45

FUNCTIONS

Autolevel •oll

TRIGGER BIND

TIME (S)

1.5

F 0.5

oCo •ec

TRIGGER BIND

oCo • Play / Sto

TRIGGER BIND

oCo • Home

TRIGGER BIND

oCo • Mark (all tracks)

TRIGGER M

Lens • Tilter

TRIGGER BIND

Lens • Rider

TRIGGER BIND

REXY FOCUS — LENS & BODY

FOCUS UNITS

M / CM

FT / IN

CAMERA BODY / FILMBACK

1x Digital Film

LENS

24-70mm Z•m f/2.8

+ ADD CUSTOM LENS

LENS BOX

3 LENSES

A primes (imported)

new box name

+ CREATE LENS BOX

+ Add lens to box...

SAVE

EXPORT

IMPORT DELETE

25mm 25mm prime

X

30mm 30mm Prime f/1.4

X

50mm 50mm Prime f/1.8

X

WIDER BIND

TIGHTER BIND

FAVOURITE LENSES

+ Add favourite...

— none yet —

FOCUS

56" 10" INV BIND

S 1.0 C 1.5 D 0 F 0

IRIS F/8.7 INV BIND

S 1.0 C 1.5 D 0 F 0

ZOOM 72MM INV BIND

S 1.0 C 1.5 D 0 F 0

REXY GRIP

CRANE

DOLLY

DRONE

Crane Yaw /rexy/craneYaw

+0.0°

live —

BIND INV RESET

S 1. C 1. D 0 F 0
MIN (°) -180 MAX (°) 180

Crane Scope /rexy/scope

5.50m

live —

BIND INV RESET

S 1. C 1. D 0 F 0
MIN (CM) 100 MAX (CM) 1000

Crane Pitch /rexy/cranePitch

+0.0°

live —

BIND INV RESET

S 1. C 1. D 0 F 0
MIN (°) -45 MAX (°) 45

CRANE TARGET (CAMERA) X

—

set

Y

—

set

Z

—

set

SET

ZERO

Core concepts (read this first)

Binding. Almost every control has a **Bind** button. Click it, then actuate the input you want — a wheel, a stick, a keyboard key, a MIDI knob, or an OSC/FreeD channel. For continuous controls, push it the way you want to be "positive" first; for keys/buttons it captures a two-stage *increase* then *decrease*. Right-click a Bind button to clear it.

Per-camera bindings. Every camera you select keeps its **own** set of bindings. Switch cameras and your inputs re-map to that camera's saved layout.

S / C / D / F tuning. Wherever you see these four little boxes, they shape how an input drives the value:

- **S — Sensitivity / Speed.** Multiplies how fast the control moves the value. Works for analogue *and* keyboard/button inputs (higher = faster).
- **C — Curve.** Response gamma. `1.0` = linear; higher = "fluid-head" feel (gentle at first, faster as you push); lower = punchy.
- **D — Deadband.** Ignores tiny inputs near centre — kills stick/wheel drift. (Not relevant to keyboard/button inputs.)
- **F — Feather.** Smooths the start and stop so motion eases in and out instead of snapping.

Inv / Hold / Reset. **Inv** flips a binding's direction. **Hold** (on wheels) freezes that axis so its binding is ignored. **Reset** returns the control to centre/default.

1. Top bar



- **Version badge** (`v3.1.0-beta`) — the running build. If it's missing or wrong after an update, press **Ctrl+Shift+R**.
- **HARDWARE** dot — your gamepad/wheels. Wakes when you click the window and press a physical button.
- **WS** dot — the internal app↔bridge link.
- **UE RC** dot — the Unreal Remote Control connection.
- **Units** — global metric (m/cm) vs imperial (ft/in) for distances and positions.
- **Dark / Light** — switches the whole interface between the default dark theme and a pale blue-grey light theme. Your choice is remembered, and every accent colour stays the same in both.

- **Check for updates** — manually checks the release server for a newer build; if one exists it downloads in the background and offers **Restart to update**. Nothing is ever checked automatically on launch.

2. Output mode & network

The **osc / ue** vs **freed** toggle chooses what REXY Bridge drives, and the network fields beneath it change to match.

- **osc / ue** (default) — drives Unreal directly via Remote Control. The fields show the UE/OSC target host and port.

SCRIPT HOST	WS PORT	UES HOST	UES OSC PORT	BASE PATH	CONFIG
127.0.0.1	9000	127.0.0.1	8000	/rexy	EXPORT IMPORT

- **freed** — broadcasts a **FreeD** camera-tracking stream on the network so any FreeD-compatible app can consume it (no UE writes). The fields switch to the FreeD stream host/port.

SCRIPT HOST	WS PORT	FREED HOST	FREED PORT	CAM ID	FPS	CONFIG
127.0.0.1	9000	127.0.0.1	40000	1	30	EXPORT IMPORT

Your connection settings, along with every binding and tuning value, save and reload from the Config controls — export a `.json` to move a whole setup between machines.

3. Cameras — Scan & select

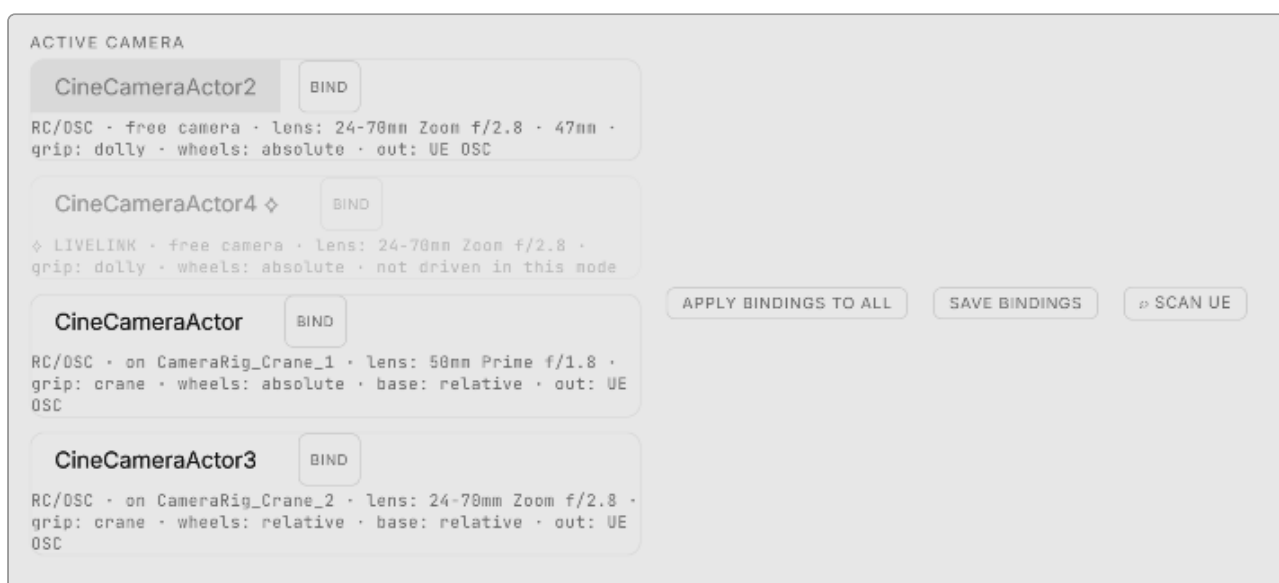
Before scanning, the camera row is empty and waiting.

ACTIVE CAMERA	APPLY BINDINGS TO ALL	SAVE BINDINGS	SCAN UE
— none found · expose a property per camera —			

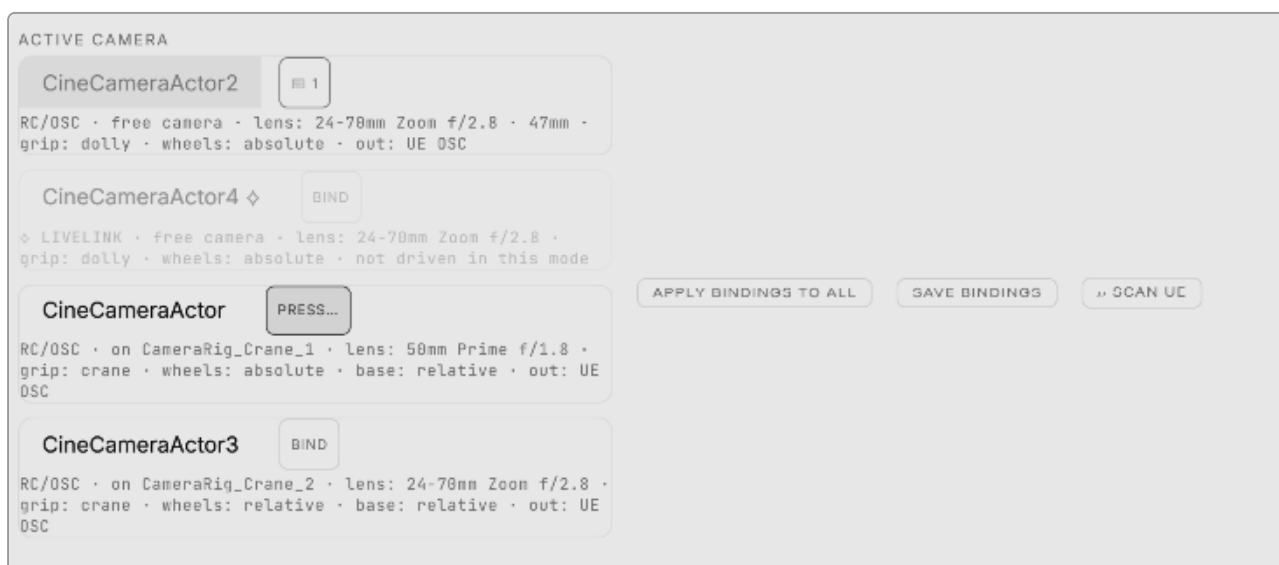
Click ▶ **Scan UE** to auto-discover every Cine Camera Actor and CameraRig_Crane in your open level — no REXYControl preset, no `mappings.json` editing. Review the result, then **Apply** for this session or **Apply + Save** to persist it.



Discovered cameras appear as buttons. Click one to make it the **active** camera; its bindings and profile load.



Each camera button can itself be **bound** to a key or gamepad button for hands-free switching mid-shot.



4. REXY Wheels — camera head

The three head axes: **pan**, **tilt**, **roll**.

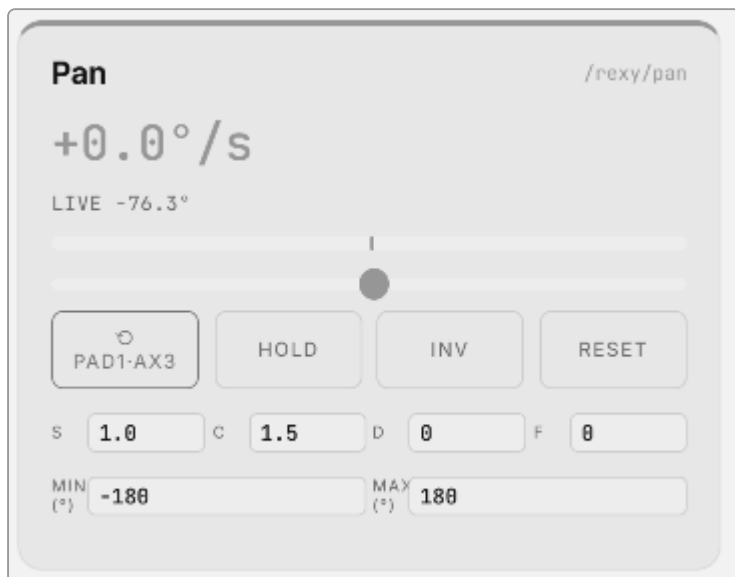
- **Absolute** — wheel position = camera angle.

The screenshot shows the 'REXY WHEELS — CAMERA HEAD' interface in 'ABSOLUTE' mode. It features three control cards for Pan, Tilt, and Roll. Each card displays a 'Value' (e.g., +180.0° for Pan) and a 'LIVE' readout (e.g., +122.5° for Pan). Below the readouts are sliders and buttons for 'PAD1-AX3', 'HOLD', 'INV', and 'RESET'. At the bottom of each card is a tuning row with 'S', 'C', 'D', and 'F' parameters, and 'MIN' and 'MAX' range settings. The 'Roll' card also includes a 'PAD1-B0 / PAD1-B2' button.

- **Relative** — wheel deflection = rotation *speed*, centre = hold (endless-rotation, fluid-head style).

The screenshot shows the same 'REXY WHEELS — CAMERA HEAD' interface but in 'RELATIVE' mode. The 'Value' readouts now show rotation speed in degrees per second (e.g., +0.0°/s for Pan). The 'LIVE' readouts show the current camera angle (e.g., +122.5° for Pan). The interface layout, including sliders, buttons, and tuning parameters, remains identical to the Absolute mode.

Each card has **Value** and **live** — readouts (commanded vs the camera's actual angle from UE), a **slider**, **Bind / Hold / Inv / Reset**, the **S / C / D / F** tuning row, and **Min / Max** to clamp the output range.



5. Functions

Bindable one-shot actions. Each card has a **Trigger** (fire it now) and a **Bind** (assign a key/button). Binding any copy of an action keeps every copy in sync.



- **Autolevel Roll** — smoothly returns roll to 0° over **Time** seconds, with an **F** easing.
- **MoCo · Rec / Play / Stop / Home / Mark** — transport actions, bindable so you never reach for the mouse mid-take. **Mark** drops a mark on **every bound track** at the playhead.
- **Lens · Tighter / Wider** — step to the next longer/shorter lens in the active **lens box**.

6. REXY Focus — lens & body

Lens & body

- **Camera body / filmback** — sensor/filmback preset.
- **Lens** — pick a lens; UE's LensSettings (focal, aperture, min-focus) update live.
- **+ Add custom lens** — define name, focal min/max, f-stop min/max, and **Min focus** (shown in cm or ft to match your Focus-units toggle). It's added to the list and applied.

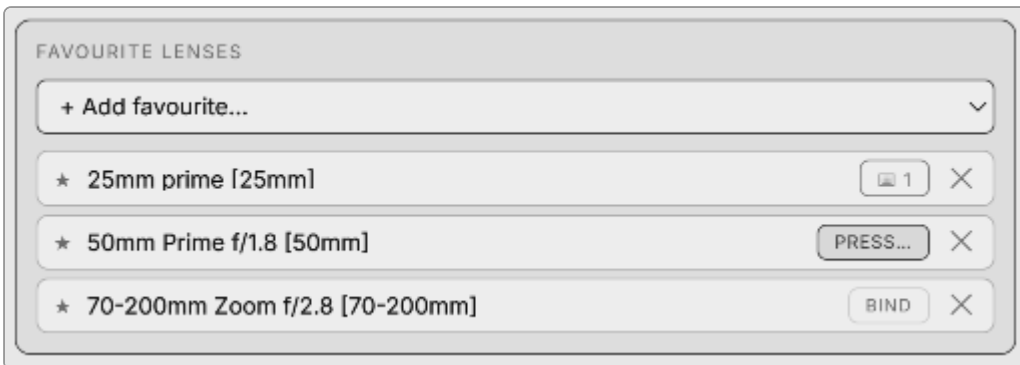
Lens boxes

A **lens box** is a named kit — a list of lenses you cycle through on set.

- **Box selector + Create lens box** — make and switch between kits.

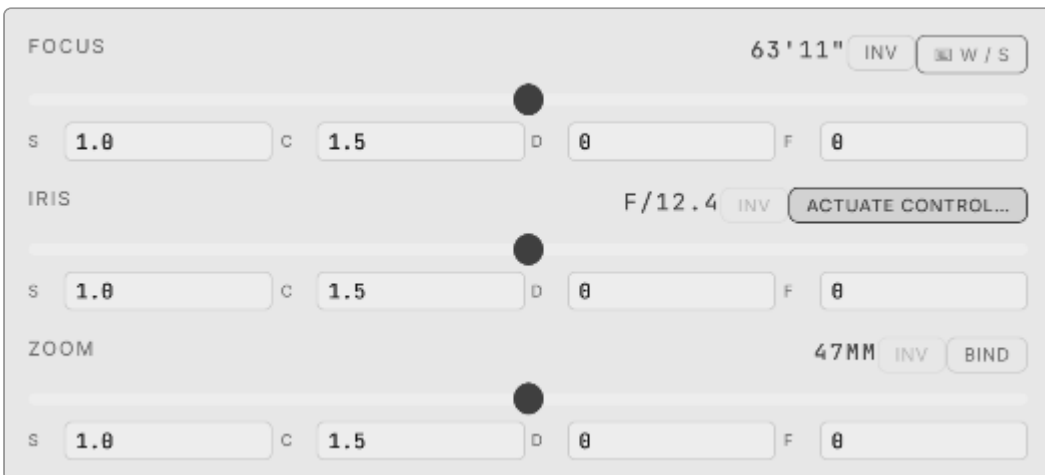
- **Add lens to box** dropdown — add any existing lens, or  **+ Custom lens...** to define a new one that drops straight into the box.
- **Lens chips** — the box's lenses, always sorted by focal length; the active lens is highlighted. **X** removes one.
- **◀ Wider / Tighter ▶** — step through the box by focal length (clamped at the ends). Each has a **Bind** for hardware control.
- **Save / Export / Import / Delete** — boxes export and import independently of the main config, so you can share a kit.

Favourite lenses



A global quick-jump list. Add any lens, then **Bind** a key/button to jump straight to it — independent of which box is loaded.

Focus / Iris / Zoom



Each has a slider, **Bind**, **Inv**, a live readout, and the full **S / C / D / F** tuning row — so a focus puller can bind a wheel or keys and tune the feel exactly like the head axes. Zoom locks automatically on a prime lens.

7. Grip — crane / dolly / drone

REXY GRIP

CRANE DOLLY DRONE

Crane Yaw

/rexy/craneYaw

+0.0°

LIVE X +0cm

BIND

INV

RESET

S 1.0 C 1.5 D 0 F 0

MIN (°) -180 MAX (°) 180

Scope

/rexy/scope

5.50m

LIVE Y +0cm

BIND

INV

RESET

S 1.0 C 1.5 D 0 F 0

MIN (CM) 100 MAX (CM) 1000

Crane Pitch

/rexy/cranePitch

+0.0°

LIVE Z +0cm

BIND

INV

RESET

S 1.0 C 1.5 D 0 F 0

MIN (°) -45 MAX (°) 45

CRANE TARGET (CAMERA) X 0.0 set Y 0.0 set Z 0.3 set SET ZERO

CRANE BASE

ABSOLUTE RELATIVE

Base X

/rexy/baseX

+0.00 m/s

LIVE X -44.04m

BIND

INV

RESET

S 1.0 C 1.5 D 0 F 0

MIN (CM) -1200 MAX (CM) 200

Base Y

/rexy/baseY

+0.00 m/s

LIVE Y +11.67m

BIND

INV

RESET

S 1.0 C 1.5 D 0 F 0

MIN (CM) -750 MAX (CM) 270

Base Z

/rexy/baseZ

+0.00 m/s

LIVE Z +3.57m

BIND

INV

RESET

S 1.0 C 1.5 D 0 F 0

MIN (CM) 0 MAX (CM) 1000

CRANE BASE (RIG) X -4404.3 set Y 1167.3 set Z 357.0 set SET ZERO

- **Crane** — the grip swings a jib arm around its base (yaw, pitch, scope/arm-length).
- **Dolly** — straight ground translation (X/Y/Z), no arm pivot.
- **Drone** — free flight through 3D space.

The grip cards (craneYaw, scope, cranePitch) use the same binding + S/C/D/F model as the wheels.

Crane Base

The base (baseX/Y/Z) drives the rig's pivot, with two modes:

- **Absolute** — slider position = base position between Min and Max.

REXY GRIP

CRANE DOLLY DRONE

X · left / right

/rexy/craneYaw

+0cm

LIVE X +0cm

BIND

INV

RESET

S

1.0

C

1.5

D

0

F

0

MIN

(°)

-180

MAX

(°)

180

Y · fwd / back

/rexy/scope

+0cm

LIVE Y +0cm

BIND

INV

RESET

S

1.0

C

1.5

D

0

F

0

MIN

(CM)

100

MAX

(CM)

1000

Z · up / down

/rexy/cranePitch

+0cm

LIVE Z +0cm

BIND

INV

RESET

S

1.0

C

1.5

D

0

F

0

MIN

(°)

-45

MAX

(°)

45

DOLLY TARGET (CAMERA)

X

0.0

set

Y

0.0

set

Z

0.0

set

SET

ZERO

- **Relative** — slider deviation = movement speed (m/s); centre = stop.

CRANE BASE

ABSOLUTE RELATIVE

Base X

/rexy/baseX

+0cm

LIVE X -44.04m

BIND

INV

RESET

S

1.0

C

1.5

D

0

F

0

MIN

(CM)

-1200

MAX

(CM)

-200

Base Y

/rexy/baseY

+0cm

LIVE Y +11.67m

BIND

INV

RESET

S

1.0

C

1.5

D

0

F

0

MIN

(CM)

-730

MAX

(CM)

270

Base Z

/rexy/baseZ

5.00m

LIVE Z +3.57m

BIND

INV

RESET

S

1.0

C

1.5

D

0

F

0

MIN

(CM)

0

MAX

(CM)

1000

8. Position panels

Live **X / Y / Z** readouts, with **Set** (send a manual position; blank axes keep their current value) and **Zero** (reset to 0,0,0).

- **Crane target** — where the camera/target sits.

CRANE TARGET (CAMERA)

X

0.0

set

Y

0.0

set

Z

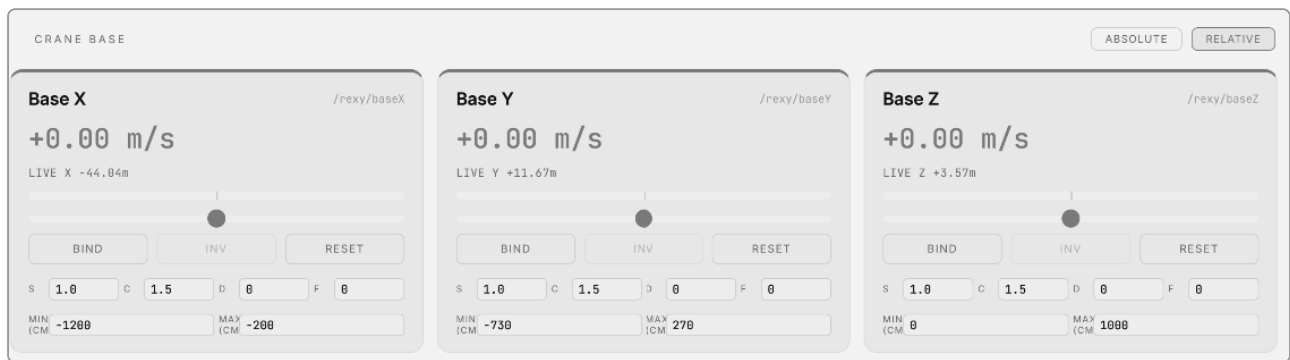
0.0

set

SET

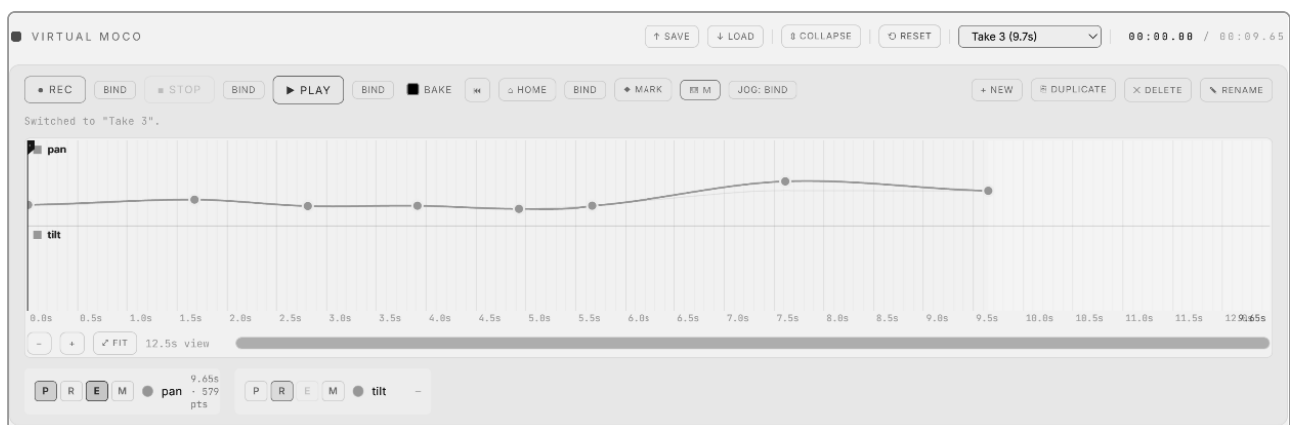
ZERO

- **Crane base** — where the rig base sits.

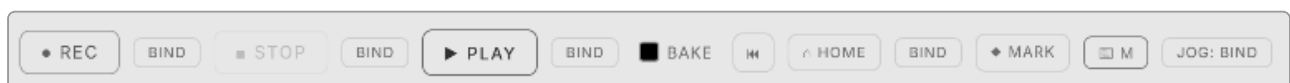


9. Virtual MoCo

Record, edit, and play back moves across every bound parameter.



Transport



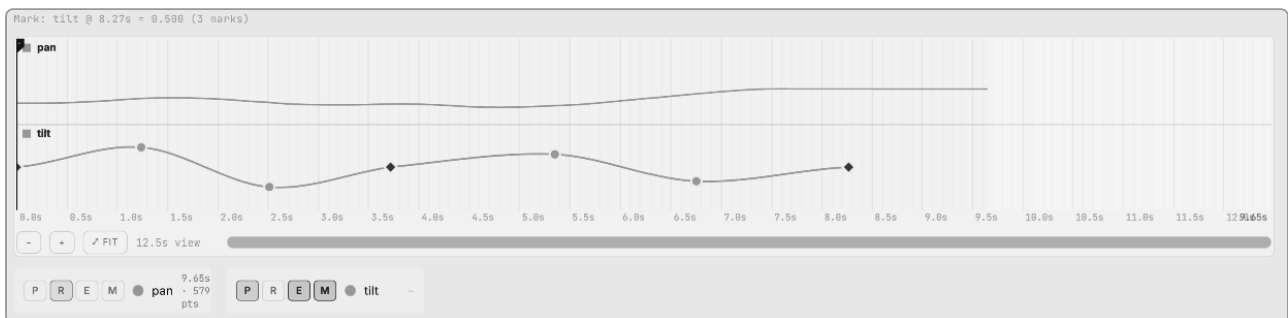
- **● REC / ■ STOP / ► PLAY** — each with a **Bind** button. STOP leaves the playhead where it is; **Home** (or Rewind) returns it to the start.
- **Bake** — a 3-2-1 countdown on REC/PLAY so you can sync with UE's Take Recorder.
- **△ HOME** — snaps play-armed tracks to their start values and the playhead to 0.
- **◆ MARK** — drops a mark on every bound track at the playhead.
- **Jog: Bind** — bind an analogue wheel/axis (or keys) to **shuttle** the playhead: push from centre to scrub forward/back, speed proportional to deflection, centre = hold.

Takes



Multi-take management. Save/Load moves as `.rxmove` files. **Expand** stacks each track in its own lane; **Reset** clears everything.

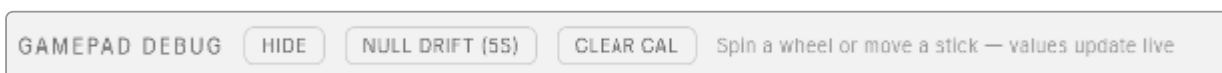
Timeline & tracks



Colour-coded per track. Scrub by dragging the playhead, zoom with the scroll wheel, and drag anchor dots to reshape curves. Each track (left of the timeline) has four arm toggles:

- **P — Play** — plays back its recorded/recorded data.
- **R — Record** — captures live input on the next REC (recording over a track auto-erases its marks).
- **E — Edit** — drag anchor dots to reshape. Available once a track has recorded data **or 2+ marks** — entering E on a marked-but-unrecorded track lays down draggable anchors that smooth the marks into **S-curves**.
- **M — Mark** — drops a fixed point at the playhead; the curve always passes through marks. Placing a mark auto-adds a start mark at $t=0$ so the move has a defined beginning.

10. Gamepad Debug & calibration



- **Show / Hide** — toggles the live view of every axis and button on connected controllers.
- **Null Drift (5s)** — samples all axes for 5 seconds and stores the average as zero, nulling resting drift. **Clear Cal** removes it.

Pad 1 - Raxy Wheels Device (Vendor: cafe Product: 4004)

10 axes - 32 buttons - mapping: (none)

ax 0 0.000 [C] ax 1 0.000 [C] ax 2 0.000 [C] ax 3 0.000 [C] ax 4 0.000 [C] ax 5 0.000 [C]

ax 7 0.000 [C] ax 8 0.000 [C] ax 9 1.23e+9 ▲

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
20	21	22	23	24	25	26	27	28	29	30	31				

The per-axis "**C**" button opens the calibration wizard. Steps 1-3 capture the axis **max**, **min**, and **rest** so uneven hardware maps to a clean -1...+1. On the live bar the **blue** marker is the captured max and the **pink** marker is the captured min.

AXIS CALIBRATION

Pad 1 - Axis 3

Step 1 of 3 — capture maximum

Move the axis to its **maximum** position, hold for a moment, then let it settle back.
Click Next when the MAX reading looks right.

raw 0.236 min - rest - max 0.291

CANCEL

REDO STEP

SKIP → TUNER

NEXT

The **Tune response** step shapes the feel, with a live waveform and curve preview. **Skip** jumps straight to the tuner; **Redo step** re-captures the current step.

AXIS CALIBRATION

Pad 1 · Axis 3

Tune response

Captured: min -0.197, rest -0.006, max 0.291. Now shape how the input responds — move it and watch the waveform.

raw 0.024min -0.197rest -0.006max 0.291

Response curve

PRESETLinear

CLOSE

BACK

SAVE

Tune response

Pick a **preset** — **Linear** (a straight 1:1 response)...

Response curve

PRESETLinear

...or **Relaxed** / **Aggressive**, or drag the curve to a custom shape. Gentle-at-centre curves give fine control near rest; steeper ends give quicker throws.



Signal Filter Lab

Rough or noisy inputs — like a load-cell Grip — can be cleaned up here, live, without touching the device firmware. **Capture** records a few seconds of the axis: the **pink** line is the raw signal, the **cyan** line is the filtered result.



+ Signal filter opens a 3×3 grid of the nine filters — Median, Moving average, EMA low-pass, Double-EMA, Kalman, 1-Euro, Hampel, Deadband and Slew limit — each previewing its effect on your captured signal, with editable parameters. **+ Add** appends a filter to the live chain, so several can stack in order and the combined result drives UE in real time.



The ⓘ icon on each filter explains what it does and what every parameter changes, so you can tune by feel rather than guesswork.

More info

Sorts a window of recent samples and takes the middle one. A lone spike can never be the median of its neighbours, so it is discarded outright, while a real edge passes through almost untouched. The best tool for saw and spikes with near-zero lag.

win - How many samples in the window (odd numbers).
Bigger = smoother, but a longer settle after a sharp reversal.

Save stores the current chain as a named **filter set** you can **Load** onto any axis or share between machines. **Scan 10s** records raw + filtered + output to a JSON log with a noise-reduction figure, for before/after comparisons.

11. Other inputs — OSC, MIDI, FreeD

FreeD in
☒ listen port off
- nothing received -

OSC in
port prefix in min in max
OSC axes behave like gamepad axes (rest = mid-range); /btn/ paths (0/1) bind two-stage like keys. Bundles supported.
- nothing received -

MIDI in
 not enabled - MIDI devices are read by THIS panel (like gamepads)
- nothing received -

- **OSC in** — accept OSC from external devices (e.g. the REXY Wheels). Set **port**, an optional **prefix** filter, and the input **min/max** range. **Sample 10s** records every OSC message for ten seconds and exports a JSON log — the ground-truth capture for diagnosing a device.
- **MIDI** — **Enable MIDI** reads MIDI controls (knobs, faders, notes, pitch-bend) into the same input path as gamepads, so they bind the same way.
- **FreeD in** — accept an incoming FreeD tracking stream.

All three feed the same binding system: enable the source, then **Bind** any control to it.

FreeD in
☒ listen port off
- nothing received -

OSC in
port prefix in min in max
OSC axes behave like gamepad axes (rest = mid-range); /btn/ paths (0/1) bind two-stage like keys. Bundles supported.
- nothing received -

MIDI in
 MIDI enabled - no inputs found yet. Plug one in.
cc6.ch1: 0.000 [0.00 .. 0.03] ~s ~ gap 41ms ~ 414 spikes Δ
cc7.ch1: 0.000 [- .. -] note0.ch1: 0.000 [- .. -] note1.ch1: 0.000 [- .. -] note2.ch1: 0.000 [- .. -] note3.ch1: 0.000 [- .. -] cc4.ch1: 0.000 [- .. -]
cc10.ch1: 0.528 [- .. -] cc8.ch1: 0.449 [- .. -] cc9.ch1: 0.000 [- .. -] cc5.ch1: 0.000 [- .. -]

12. OSC output log

OSC OUTPUT

```
Lens → 24-70mm Zoom f/2.8 (24-70mm zoom, f/2.8-22)
Grip mode → dolly
Wheel mode → continuous
Active camera → CineCameraActor_6
UE: Discovered via actor-search.
Output → UE Remote Control
```

A live feed of what the bridge is sending and status messages — handy for confirming a control is reaching UE and for spotting warnings.

Questions or bug reports: rob@realprogear.com